

Exception and exception handling

1

Your Code

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         try{
6             int num1=sc.nextInt();
7             int num2=sc.nextInt();
8             System.out.println("enter the 1st integer:"+num1);
9             System.out.println("enter the 2nd integer:"+num2);
10            int result=num1/num2;
11            System.out.println("Result:"+result);
12        }catch(ArithmeticException e){
13            System.out.println("error: cannot divided by zero");
14        }catch(Exception e){
15            System.out.println("error : invalid input");
16        }finally{
17            sc.close();
18        }
19    }
20 }
```

Input

10
0

Output

enter the 1st integer:10
enter the 2nd integer:0
error: cannot divided by zero

2

Your Code

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int arr[]={10,20,30,40,50};
6         System.out.println("array elements : 10,20,30,40,50");
7         System.out.println("enter an index(0 to 4): ");
8         try{
9             int index=sc.nextInt();
10            System.out.println("element at index " + index+"is: "+arr[index]);
11        }catch(ArrayIndexOutOfBoundsException e){
12            System.out.println("error index is out of range(0 to 4)");
13        }finally{
14            sc.close();
15        }
16    }
17 }
```

Input

5

Output

array elements : 10,20,30,40,50
enter an index(0 to 4):
error index is out of range(0 to 4)

3

Your Code

```
1 import java.util.Scanner;
2 public class NumberFormatExample {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         System.out.println("Enter a string to convert an integer: ");
6         String input = sc.nextLine();
7         try {
8             int num = Integer.parseInt(input);
9             System.out.println("Conversion successful! The integer is: " + num);
10        } catch (NumberFormatException e) {
11            System.out.println("Error: Invalid input. The provided string is not");
12        } finally {
13            sc.close();
14        }
15    }
16 }
```

Input

helo

Output

Enter a string to convert an integer:
Error: Invalid input. The provided string is not a valid integer.

Run

Save

✓ Draft saved

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         try {
6             System.out.println("Enter first integer: ");
7             int a = sc.nextInt();
8             System.out.println("Enter second integer: ");
9             int b = sc.nextInt();
10            int result = a / b;
11            int[] arr = new int[1];
12            arr[1] = result;
13            System.out.println("result: " + arr[0]);
14        } catch (ArithmeticException e) {
15            System.out.println("ArithmeticException occurred: Cannot divide by zero.");
16        } catch (ArrayIndexOutOfBoundsException e) {
17            System.out.println("ArrayIndexOutOfBoundsException occurred: Array index out of bounds.");
18        } finally {
19            sc.close();
20        }
21    }
22 }
```

10
0

Output

```
Enter first integer:
Enter second integer:
ArithmeticException occurred:
Cannot divide by zero.
```